

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Wednesday 9th May 2012

09:30 to 11:30

Year 3 Courses

Convener: K. Kalorkoti

External Examiners: K. Eder, A. Frisch, J. Gurd

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) What are the advantages and disadvantages of protocol layering? [5 marks]
(b) Describe four negative consequences of network congestion. [4 marks]
(c) Discuss two broad approaches towards congestion control, each with an example. [3 marks]
(d) Consider the effect of using slow-start on a link with a 20ms round-trip time and no congestion. The receive window is 16KB and the maximum segment size is 1460 bytes. How long does it take before the first full window can be sent in one transmission round? [2 marks]
(e) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assume *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received. Also assume approximately constant RTTs (round-trip times).
 - i. How long does it take for *CongWin* to increase from 1 MSS to 5 MSS (assuming no loss events)? [2 marks]
 - ii. What is the average throughput (in terms of MSS and RTT) for this connection over that period (from sub-part (i))? [2 marks]
- (f) In networking, max-min fairness is the classical notion of fairness. A throughput distribution among users is max-min fair if a user's throughput cannot be increased without decreasing the throughput of other users with same or lower throughput. Does TCP always provide max-min fairness? Why or Why not? [5 marks]
- (g) At what level of granularity does TCP provide fairness? Outline an application scenario where the user can exploit this fact to get a greater than fair share of the network bandwidth. [2 marks]

2. (a) Suppose we are given an Internet connection with 40Mbps bandwidth to be shared between multiple users. Suppose each user is active only 20% of the time. State how many users can be supported when using circuit switching. For packet switching, consider 3 cases, each with a different number of total users:

- i. 4 users
- ii. 6 users
- iii. 8 users

Between the above 3 cases, what is the most number of users supported by packet switching?

[7 marks]

- (b) List five key differences between IPv4 and IPv6.

[5 marks]

- (c) Discuss two mechanisms for transition from IPv4 to IPv6.

[4 marks]

- (d) Suppose a router has the following CIDR entries in its routing table:

Net/Prefix	Next Hop
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives? (i) 135.46.63.10; (ii) 135.46.57.14; (iii) 135.46.52.2; (iv) 192.53.40.7; (v) 192.53.56.7.

[5 marks]

- (e) Scalability of link state routing protocols, exemplified by OSPF, can be improved by making them hierarchical as with hierarchical OSPF. Suggest an alternative and more effective way to improve the scalability of link state protocols.

[4 marks]

3. (a) In digital subscriber line (DSL) based broadband access networks, each user's broadband modem is connected to the infrastructure of the Internet Service Provider (ISP) via a dedicated (phone) line, whereas cable modem based broadband users are connected to ISP infrastructure via shared cable. Does this mean DSL based approach is superior to the cable modem based approach? Explain. [2 marks]
- (b) Suppose the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. The corresponding sequence of hex values for those integers is: 0xf3, 0x51, 0x80, 0x40, 0x20, 0x10, 0x1b, 0x09. Compute the Internet checksum for this data. [4 marks]
- (c) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of a bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 10011101.
- i. Show the actual bit stream transmitted.
 - ii. Suppose that the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.
 - iii. Give an example of bit errors in the bit stream transmitted that will not be detected by the receiver. [6 marks]
- (d) Cryptographic hash functions (e.g., MD5) are similar to checksum and CRC, and additionally offer security related benefits. Then why are they not used for error detection in networks. [2 marks]
- (e) Give two reasons why networks might use an error-correcting code instead of error detection and retransmission. [4 marks]
- (f) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [7 marks]